

TITLE PAGE

Problem Statement ID <<FILL IN>>

Problem Statement Title Eco-Loop Building Agents — autonomous closed-loop HVAC control using EnergyPlus, MCP and a local open-source LLM

Theme <<FILL IN>> · PS Category Software

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TECH STACK

EnergyPlus 26.1 · pyenergyplus · FastMCP · Model Context Protocol · Ollama + Llama 3.2 3B · Pydantic · Streamlit

A BUILDING THAT RUNS ITSELF

EnergyPlus supplies the physics. A local open-source LLM supplies the judgement. The Model Context Protocol carries every reading and every command between them — while the simulation is still running.

–8.5%

HVAC energy
summer week

336/336

agent turns
zero failures

100% open source

One laptop — no cloud, no API keys, no per-query cost

IDEA TITLE

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PROPOSED SOLUTION

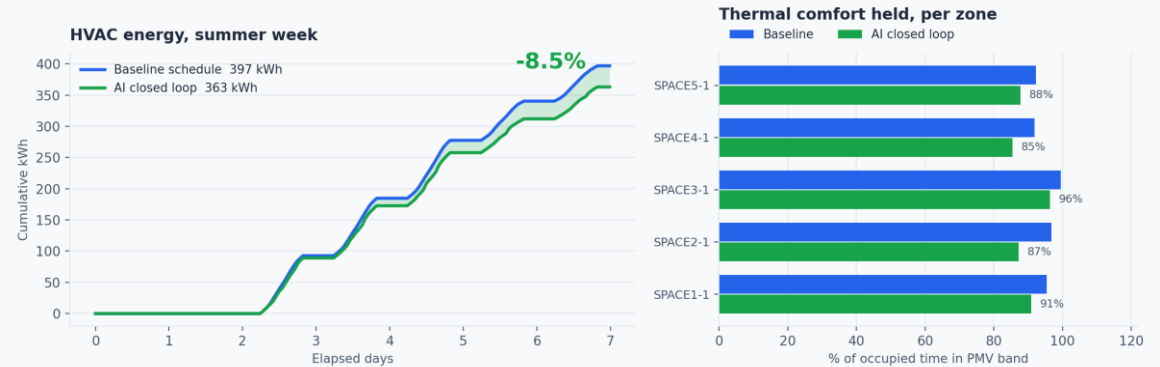
Traditional BMS follow rigid clock schedules. Eco-Loop turns the building into a self-correcting agent.

HOW IT WORKS

- EnergyPlus runs in-process; sensors read every zone timestep
- Every 60 simulated minutes the state reaches a local LLM via six MCP tools
- The model weighs comfort, demand and grid carbon, then returns setpoints

INNOVATION

- Injected into the LIVE solver — no IDF rewrite, no restart
- MCP is the safety boundary: every command validated server-side



–8.5% HVAC energy

Measured, not projected — while holding occupants closer to thermal neutrality than the baseline (mean PMV $-0.43 \rightarrow -0.02$)

WHY IT IS DIFFERENT

Most LLM-and-simulation work rewrites the model file and re-runs it — that can only compare finished runs. It cannot react to a zone drifting out of comfort at 14:00 on day 3.

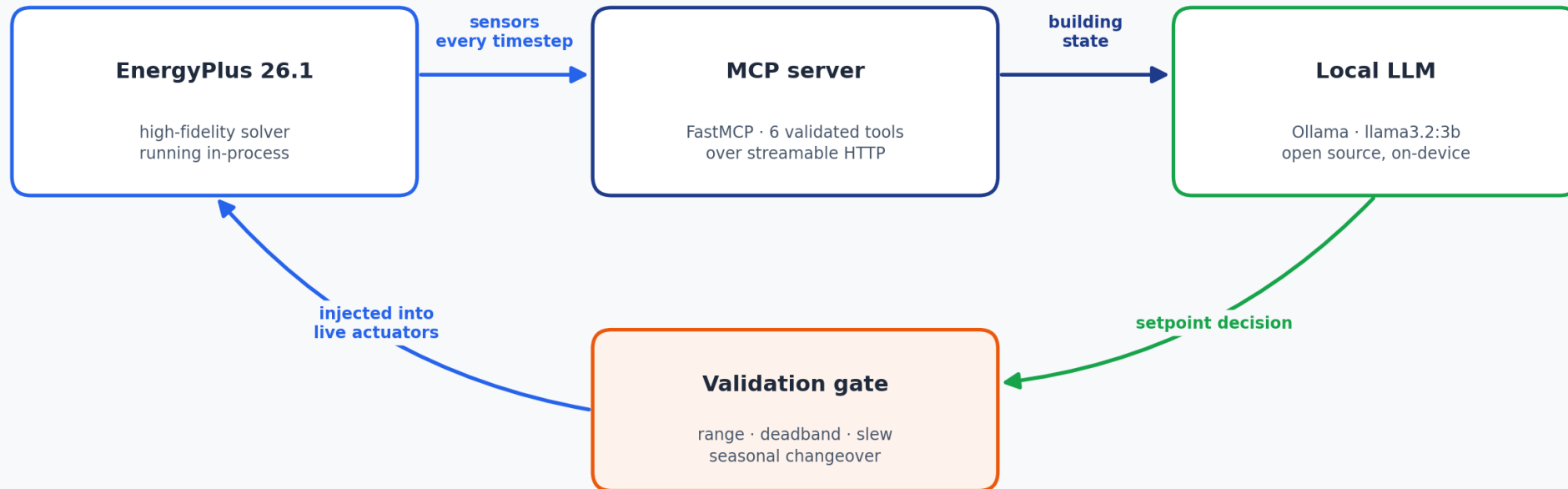
TECHNICAL APPROACH

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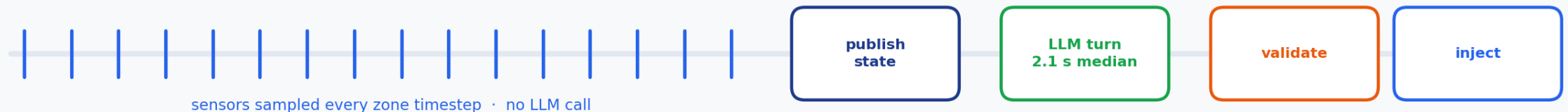
TECHNOLOGIES EnergyPlus 26.1 (pyenergyplus C API) · Python · FastMCP over streamable HTTP · Ollama running Llama 3.2 3B · Pydantic · Streamlit + Plotly

METHODOLOGY sample every timestep → aggregate per control interval → LLM tool-calling turn → validate → inject into live actuators

CLOSED LOOP · no file rewrite, no restart · 168 agent decisions per simulated week



ONE CONTROL INTERVAL · 60 SIMULATED MINUTES



FEASIBILITY AND VIABILITY

336 / 336

agent turns — 0 fallbacks, 0 timeouts, 0 actuation errors

−8.5%

HVAC energy, summer week (−3.7% winter)

2.1 s

median agent decision latency, fully on-device

RISKS & CONSTRAINTS

- LLM latency inside a blocking simulation callback
- A small model can move the wrong setpoint, or run heating against cooling
- One setpoint pair cannot satisfy five differently-loaded zones

HOW EACH IS SOLVED

- Hard per-turn deadline; on timeout the loop holds the last accepted command
- Observations state the required direction — the model cannot invert physics
- Seasonal changeover parks the idle setpoint; a per-zone trim corrects each zone

VIABILITY AT SCALE Buildings are ~40% of global energy — a single-digit HVAC reduction is material at portfolio scale. The loop is model-agnostic, and a larger cognitive engine needs no change to the MCP tool layer.

EVERY COMMAND PASSES FIVE GATES BEFORE IT REACHES AN ACTUATOR



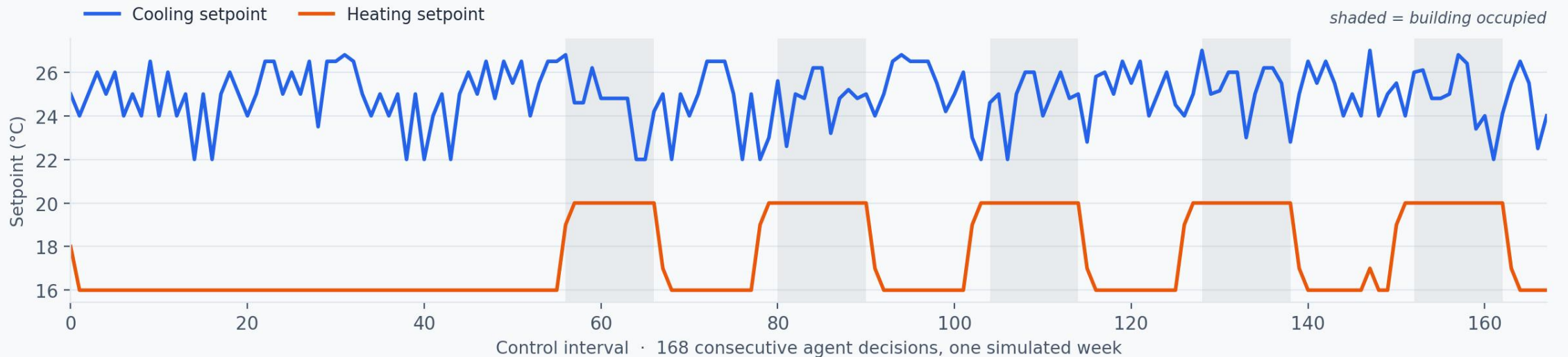
Rejected commands return accepted: false with a reason — the previous setpoints stay in force and the model can retry.

ARTIFACTS

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github.com/ayushYadav1107/Eco_loop

Baseline and runtime-generated .idf models, the full agent decision trail (agent_decisions.jsonl), per-timestep EnergyPlus results and the savings dashboard are all committed — a fresh clone reproduces every figure here.



The agent's own stated reasoning, taken verbatim from agent_decisions.jsonl

- 07-17 09:00** *"Raise cooling_sp to about 27.0 C this turn — this fixes comfort AND saves energy"*
- 07-17 11:00** *"HOLD the current setpoints. Do not trim energy here — you would push occupants out of the band"*
- 07-17 12:00** *"Nudge cooling_sp up to save energy while keeping PMV within the band"*

RESEARCH AND REFERENCES

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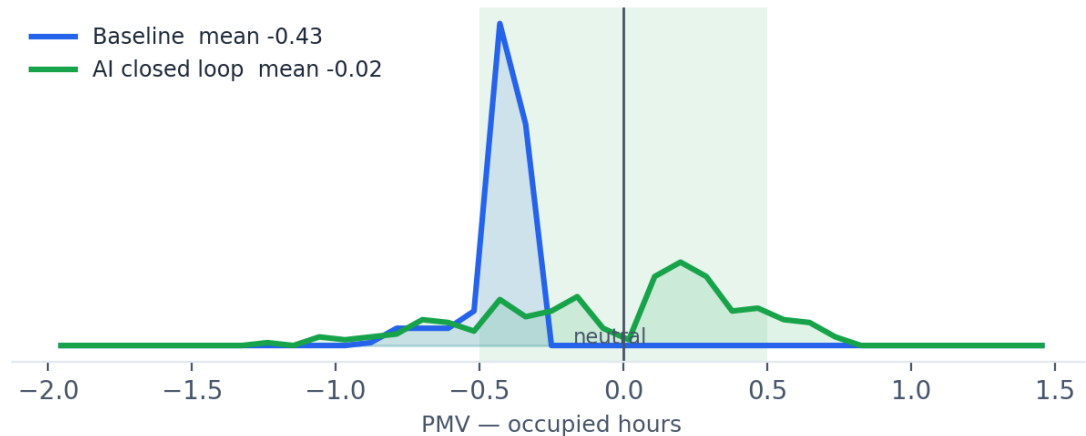
SIMULATION & BUILDING PHYSICS

- EnergyPlus Engineering Reference · EMS Application Guide
- pyenergyplus Data Exchange API — `get_variable_handle`, `set_actuator_value`
- ISO 7730:2005 · ASHRAE Standard 55 — PMV/PPD comfort model

AGENT, PROTOCOL & INFERENCE

- Model Context Protocol — `modelcontextprotocol.io` · FastMCP — `gofastmcp.com`
- Meta Llama 3.2 served locally through Ollama — `ollama.com`

Comfort was not traded away



REPRODUCE THE RESULT

```
python main.py prepare
python main.py run-baseline --start 07-15 --end 07-21
python main.py run-ai --start 07-15 --end 07-21
python main.py dashboard
```

AUDITABLE, NOT ANECDOTAL

The controller runs at temperature 0, so repeated runs return identical totals. Anyone can clone the repo and reproduce these exact figures.

HONEST LIMITATION

One setpoint pair drives five zones — per zone the agent holds 85–96% of occupied time in band. Per-zone setpoints are next; the actuator handles are already in place.